



Bayonetta v Mortal Kombat

Bayonetta creator Hideki Kamiya reveled that Mortal Kombat and Bayonetta will get a crossover. <https://digit.in/mar19-60>



No more Video on Steam

The non-gaming video section of Steam is unsurprisingly removed from the Steam store. <https://digit.in/mar19-61>

ing this counterweight to the Earth would act as the elevator. Till recently, scientists did not know of any material that was strong enough and light enough to form the cable. Recently, some materials have been discovered with the required characteristics, including diamond nanothreads, carbon nanotubes and graphene. Obayashi Corporation, a Japanese company believes that by the 2030s, the production of carbon nanotubes can be scaled to the required amount, and hopes to construct a working space elevator by the year 2050.

MIND UPLOADING

The concept of uploading a human brain to a computer is explored in *Transcendence*, *Black Mirror* and *The Outer Limits*. The video games, *Portal* and *Halo* also has this concept. Uploading a mind to a computer allows a person to become virtually immortal, but there are serious ethical considerations. These questions include if the uploaded mind is still the person, is the destruction of the copy an act of murder, and what should be done about botched or corrupted uploads. In technical parlance, uploading a brain is called whole brain emulation (WBE). Before the mind can be uploaded into a computer, researchers must create ways in which the brain can talk with a machine. This is a field of active research, with many companies around the world creating sophisticated brain-computer interfaces (BCIs). There are already neuroprosthetics in the market, that can allow people to hear or move wheelchairs. Another area of active research that can help humans eventually upload a brain, is the development of quantum computers. We will need supercomputers smarter than the human brain, in order to store it. Humans also need to understand the structure of the brain, and figure out how all the neurons connect to each other. The Human Connectome Project is dedicated to do just this, and map out the human brain. Carboncopies is a foundation that exists to accelerate

research towards mind uploading and hosts conferences towards this goal.

BIOFABRIC

The concept of biofabric was introduced in the 1970 novel, *Say Goodbye to the Wind* by JG Ballard. The author explains it as, "a dress can change its color and texture in a few seconds, becoming more decollete at the approach of an eager admirer, more formal at a chance meeting with a bank manager. This sensitivity to mood explains the real popularity of bio-fabrics. Clothes are no longer made from dead fibers of fixed color and texture that can approximate only crudely to the vagrant human figure, but from living tissues that adapt themselves to the contours and personality of the wearer." In 2017, researchers from MIT developed clothes that used liv-

CLOAKING DEVICE

Cloaking devices are common in science fiction stories set in space, and allow spacecraft to stealthily pass through hostile territories, or sneak up on enemies. The phrase was first used in the *Star Trek* episode called "The Enterprise Incident", and the technology is seen in *Star Wars*, *Doctor Who*, *Babylon 5* and *Battlestar Galactica* as well. Essentially, the cloaking device on a spaceship allows it to act like a submarine in the seas. In the real world, there are stealth aircraft and several approaches for camouflaging warships and planes. These approaches to cloaking devices involves making vehicles partially invisible in portions of the electromagnetic spectrum. BAE systems has developed a camouflaging system called ADAPTIV, which hides vehicles from thermal imaging



The microbe activated ventilation flaps on the breathing dress

ing microbial cells as actuators - tiny machines that caused the clothes to respond to certain situations. The clothes were designed for workouts, with a variety of different sized flaps arranged according to the heat maps of the human body. The cells would get activated when the wearer would perspire, and as a result, the flaps would open up, letting in more air. The researchers also developed running shoes with the same cells, with flaps to air out any build up of moisture. The cells used were benign, and even safe to consume.

systems, or infrared cameras. This is achieved by layering the vehicle with hexagonal cells or pixels, each of which can individually be rapidly heated or cooled. This allows the vehicle to mimic the ambient temperatures in its surroundings, and disguising itself from thermal imaging systems. As the panels can be individually heated, the vehicle can form a picture, that makes it look like something else entirely. A tank for example, can look like a cow, a car or rocks when seen through infrared cameras. The panels can also be used in aircraft and ships. 

Credit: Hannah Cohen/MIT